

ServiceNow Internship Program

Capstone Assessment – Gate Check 2

Scenario

Your organization NovaTech IT Services supports thousands of employee devices across global offices.

Employees frequently report issues related to laptops, tablets, and mobile devices. Currently, device information exists in external inventory systems and device support requests are logged manually.

You have been asked to build a Device Service Management system that:

1. Tracks device service requests
2. Validates device ownership
3. Automatically assigns technicians
4. Imports device inventory from external systems
5. Provides automated notifications and workflows
6. Performs data validation during import

Part 1 — Table Design and Form Configuration

Create a table: Device Service Request (extends Task)

Fields:

Employee (Reference → User)

Device ID (String)

Device Type (Choice)

Issue Category (Choice)

Issue Description (String)

Urgency (Choice)

Priority (Choice)

Assigned Technician (Reference → User)

Service Status (Choice)

Repair Cost (Currency)

Warranty Status (Choice)

Reported Date (DateTime)

Resolution Notes (String)

Tasks:

- Configure form sections (Device Information, Issue Details, Repair Details)
- Create two additional form views (Employee View, Technician View)
- Create two form templates (Laptop Screen Issue, Battery Replacement)

Part 2 — Client Scripts

Client Script 1 — Device Validation

When Device ID changes call a client callable Script Include to verify the device exists in the Device Inventory table.

Client Script 2 — Conditional Field Visibility

If Issue Category = Hardware → Show Repair Cost field otherwise hide it.

Client Script 3 — Submission Validation

Prevent submission if:

- Issue description length < 15 characters
- Device ID is empty

Part 3 — UI Policies

Policy 1:

If Device Type = Mobile → Hide Repair Cost

Policy 2:

If Service Status = Resolved → Make Resolution Notes and Repair Cost mandatory

Part 4 — Business Rules

BR1 — Priority Calculation

Calculate priority automatically based on urgency.

BR2 — Technician Assignment

Laptop → Laptop Support

Mobile → Mobile Support

Tablet → Device Support

BR3 — Status Audit

Add work note when status changes:

Status changed from <old> to <new> by <user>

BR4 — Cost Automation

Call Script Include to calculate repair cost.

Part 5 — Script Include Utility Class

Create Script Include: DeviceServiceUtils

Functions:

- Calculate repair cost based on device type
- Validate device exists in Device Inventory
- Check warranty expiry

Part 6 — Client Callable Script Include

Create client callable Script Include using AbstractAjaxProcessor called DeviceSupportHelper.

Methods:

- Return troubleshooting steps based on issue category
- Return warranty status from inventory table

Must be called from Client Script using GlideAjax.

Part 7 — Flow Designer Automation

Flow Name: Device Service Lifecycle

Trigger: Record created

Steps:

1. Validate device
2. Assign technician
3. Notify technician
4. Wait until status becomes In Progress
5. When resolved trigger completion event
6. Notify employee

Part 8 — Events and Notifications

Event: device.repair.completed

Trigger when Service Status becomes Resolved.

Notification should send:

Device ID

Repair Cost

Technician

Resolution Notes

Part 9 — Device Inventory Import

Create table Device Inventory.

Fields:

Device ID

Device Type

Model

Purchase Date

Warranty Expiry

Assigned User

Create Data Source, Import Set, and Transform Map for CSV inventory import.

Part 10 — Transform Map Scripts

OnStart: Log import start

OnBefore: Skip record if Device ID empty

Field Script: Convert codes

LT → Laptop

MB → Mobile

TB → Tablet

OnForeignInsert: Create referenced user if not found

OnAfter: If warranty expired set Warranty Status = Expired

Part 11 — Advanced Validation

Create Script Include DeviceImportValidator.

Validation rules:

- Device ID must be unique
- Purchase date cannot be future
- Warranty expiry must be after purchase date
- Device type must match allowed values

Log errors

Deliverables

Interns must submit screenshots and evidence of:

- Table configuration
- Client scripts
- UI policies
- Business rules
- Script includes
- Flow designer configuration
- Events and notifications
- Transform map configuration
- Working example records

All screenshots must be compiled in one document.

Evaluation Criteria (Weightage)

Table Design – 10

Client Scripts – 15

UI Policies – 10

Business Rules – 15

Script Includes – 15

Flow Designer – 10

Events & Notifications – 10

Transform Maps – 15